



- If $f(x) = x^n$, then $f(1) + \frac{f'(1)}{1!} + \frac{f''(2)}{2!} \dots \dots \frac{f^{(n)}(1)}{n!}$ is equal to
NIMCET 2007
(a) 0 (b) 2^n (c) 2^{n-1} (d) $\frac{n(n+1)}{2}$
- If $f'(x) = g(x)$ and $g'(x) = -f(x)$ for all x and $f(3) = 5 = f'(3)$ then $f''(19) + g''(19)$ is :
NIMCET 2007
(a) 16 (b) 32 (c) 64 (d) None
- If $u = \sec^{-1}\left(\frac{x+1}{x-1}\right) + \sin^{-1}\left(\frac{x-1}{x+1}\right)$, $x \in (0, \infty)$ and $x \neq 1$, then $\frac{dy}{dx}$ is equal to
NIMCET 2008
(a) 1 (b) $\frac{x-1}{x+1}$ (c) 0 (d) $\frac{x+1}{x-1}$
- If $y = f(x)$ is an odd and differentiable function defined on $(-\infty, \infty)$ such that $f'(3) = -2$, then $f'(-3)$ is equal to
NIMCET 2009
(a) 4 (b) 2 (c) -2 (d) 0
- Find $\frac{d}{dx}\left(\sqrt{x} - \frac{5}{\sqrt{x}}\right)$
NIMCET 2011
(a) $\frac{1}{2\sqrt{x}} + \frac{5}{2}x^{-3/2}$ (b) $2x - \frac{5}{2}x^{3/2}$
(c) $2x + \frac{5}{2}x^{-3/2}$ (d) None of these
- If $f(x) = \tan^{-1}\left[\frac{\sin x}{1+\cos x}\right]$, then what is the first derivative of $f(x)$?
NIMCET 2013
(a) $\frac{1}{2}$ (b) $-\frac{1}{2}$
(c) 2 (d) -2
- If $x = a \cos t$, $y = b \sin t$, then $\frac{d^2y}{dx^2}$ is
NIMCET 2013
(a) $-\frac{b^4}{a^2y^3}$ (b) $\frac{b^4}{a^2x^3}$
(c) $\frac{b}{ay^4}$ (d) $-\frac{a^4}{bx^3}$
- With the usual notation, $\frac{d^2x}{dy^2}$ is
NIMCET 2015
(a) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (b) $\frac{d^2y}{dy^2}\left(\frac{dy}{dx}\right)^{-2}$
(c) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-2}$ (d) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$
- If $y = \cos^2 x^2$, find $\frac{dy}{dx}$
NIMCET 2017
(a) $4x^2 \sin x^2 \cos x^2$
(b) $-4x^2 \cos x^2 \sin x^2$
(c) $2x \sin x^2 \cos x^2$
(d) $-2x \cos x^2 \sin x^2$

- The derivative of $(x^3 + e^x + 3^x + \cot x)$ with respect to x is
NIMCET 2017
(a) $3x^2 + e^x + 3^x(\log 3) - \operatorname{cosec}^2 x$
(b) $3x^2 + e^x + 3^x(\log 3) + \operatorname{cosec}^2 x$
(c) $3x^2 + e^x + 3^x(\log 3) - \sec^2 x$
(d) $3x^2 + e^x + 3^x(\log 3) + \sec^2 x$
- Differentiate $[-\log(\log x), x > 1]$ with respect to x .
NIMCET 2017
(a) $-\frac{1}{(x \log x)}$ (b) $\frac{1}{\log x}$
(c) $1/x$ (d) $x \log x$
- $\frac{d^2y}{dx^2}$ equals
NIMCET 2018
(a) $\left(\frac{d^2y}{dx^2}\right)^{-1}$ (b) $-\left(\frac{d^2y}{dx^2}\right)^{-1}\left(\frac{dy}{dx}\right)^{-3}$
(c) $\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-2}$ (d) $-\left(\frac{d^2y}{dx^2}\right)\left(\frac{dy}{dx}\right)^{-3}$
- Differential coefficient of $\log_{10} x$ w.r.t $\log_x 10$ is
NIMCET 2018
(a) $-\frac{(\log x)^2}{(\log 10)^2}$ (b) $\frac{(\log_{10} x)^2}{(\log 10)^2}$
(c) $\frac{(\log_x 10)^2}{(\log 10)^2}$ (d) $-\frac{(\log 10)^2}{(\log x)^2}$
- If $y = \tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right)$, $-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}}$, then $\frac{dy}{dx}$ is
NIMCET 2021
(a) $-\frac{1}{1+x^2}$ (b) $\frac{3}{1+x^2}$
(c) $\frac{3}{\sqrt{1+x^2}}$ (d) $\frac{1}{\sqrt{1+x^2}}$
- If $y = \sin^{-1}\left(\frac{x^2+1}{\sqrt{1+3x^2+x^4}}\right)$ ($x > 0$), then $\frac{dy}{dx}$ is equals
NIMCET 2021
(a) $\frac{x^2-1}{x^4+3x^2+1}$ (b) $\frac{x^2+1}{x^4+3x^2+1}$
(c) $\frac{x^2-1}{x^4-3x^2+1}$ (d) $\frac{x^2+1}{x^4-3x^2+1}$
- If $x^m y^n = (x+y)^{m+n}$, then $\frac{dy}{dx}$ is
NIMCET 2022
(a) $\frac{x+y}{xy}$ (b) xy
(c) x/y (d) y/x

ANSWER KEY

1.	(b)	2.	(d)	3.	(c)	4.	(c)	5.	(d)
6.	(a)	7.	(a)	8.	(d)	9.	(*)	10.	(a)
11.	(a)	12.	(d)	13.	(a)	14.	(b)	15.	(a)
16.	(d)								



Questions Asked in NIMCET 2007 to 2025

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	02	NIMCET 2016	00
NIMCET 2008	01	NIMCET 2017	03
NIMCET 2009	01	NIMCET 2018	02
NIMCET 2010	00	NIMCET 2019	00
NIMCET 2011	01	NIMCET 2020	00
NIMCET 2012	00	NIMCET 2021	02
NIMCET 2013	02	NIMCET 2022	01
NIMCET 2014	00	NIMCET 2023	00
NIMCET 2015	01	NIMCET 2024	00
		NIMCET 2025	00

SOLUTION

1. (b) $f(x) = x^n$ then $f(1) + \frac{f'(1)}{1!} + \frac{f''(1)}{2!} + \dots + \frac{f^{(n)}(1)}{n!}$

$$f(x) = x^n$$

$$f(1) = 1$$

$$f'(x) = nx^{n-1}$$

$$f'(1) = n$$

$$f''(x) = n(n-1)x^{n-2}$$

$$f''(1) = n(n-1)$$

$$f^{(n)}(x) = n(n-1)(n-2)\dots 1$$

$$f^{(n)}(1) = n!$$

$$= f(1) + \frac{f'(1)}{1!} + \frac{f''(1)}{2!} + \dots + \frac{f^{(n)}(1)}{n!}$$

$$= 1 + \frac{n}{1!} + \frac{n(n-1)}{2!} + \frac{n(n-1)(n-2)}{3!} + \dots + \frac{n!}{n!}$$

$$= {}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n$$

$$= 2^n$$

2. (d)

$$f'(x) = g(x)$$

$$g'(x) = -f(x)$$

$$f(3) = 5 = f'(3), \text{ then } f''(19) + g''(19)$$

$$\text{Let } h(x) = f''(x) + g''(x)$$

$$h'(x) = 2f(x) \cdot f'(x) + 2g(x) \cdot g'(x)$$

$$f'(x) = 0$$

It means $h(x)$ is a constant function.

$$f''(x) + g''(x) = \text{constant}$$

$$f''(3) + g''(3) = 5^2 + 5^2 = 50$$

$$f''(19) + g''(19) = 50$$

3. (c) $y = \sec^{-1}\left(\frac{x+1}{x-1}\right) + \sin^{-1}\left(\frac{x-1}{x+1}\right), x \in (0, \infty)$

$$y = \cos^{-1}\left(\frac{x-1}{x+1}\right) + \sin^{-1}\left(\frac{x-1}{x+1}\right)$$

$$\therefore y = \frac{\pi}{2}$$

$$\left[\because \sin^{-1}x + \cos^{-1}x = \frac{\pi}{2} \right] \Rightarrow \frac{dy}{dx} = 0$$

4. (c) $y = f(x)$

$$f(-x) = -f(x) \text{ [As } f(x) \text{ is odd function]}$$

$$\Rightarrow -f'(-x) = -[f'(x)]$$

$$\Rightarrow f'(-x) = [f'(x)]$$

Hence, $f'(x)$ will be an even function.

$$f'(3) = f'(-3) = -2$$

$$5. (d) \frac{d}{dx} \left[\sqrt{x} - \frac{5}{\sqrt{x}} \right] = \frac{1}{2\sqrt{x}} - 5 \frac{d}{dx} (x)^{-1/2}$$

$$= \frac{1}{2} (x)^{-1/2} + \frac{5}{2} x^{-3/2}$$

$$6. (a) f(x) = \tan^{-1} \left[\frac{\sin x}{1 + \cos x} \right] = \tan^{-1} \left[\frac{2 \sin \frac{x}{2} \cos \frac{x}{2}}{2 \cos^2 \frac{x}{2}} \right]$$

$$f(x) = \tan^{-1} \left[\tan \left(\frac{x}{2} \right) \right] = \frac{x}{2} \Rightarrow f'(x) = \frac{1}{2}$$

7. (a) $x = a \cos t, y = b \sin t$

$$\frac{dx}{dt} = -a \sin t, \frac{dy}{dt} = b \cos t \Rightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = -\frac{b}{a} \cot(t)$$

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{d}{dt} \left(\frac{dy}{dx} \right) \frac{dt}{dx} = \left(-\frac{b}{a} \right) (-\operatorname{cosec}^2 t) \left(\frac{dt}{dx} \right)$$

$$= \frac{b}{a} (\operatorname{cosec}^2 t) \left(-\frac{1}{a \sin t} \right)$$

$$= -\left(\frac{b}{a^2} \right) \operatorname{cosec}^3 t = -\frac{b}{a^2 \sin^3 t}$$

$$= -\frac{b^4}{a^2 y^3} \quad (\text{as } \sin t = \frac{y}{b})$$

8. (d) Since, $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}} = \left(\frac{dy}{dx} \right)^{-1}$

$$\Rightarrow \frac{d^2x}{dy^2} = \frac{d}{dy} \left(\frac{dx}{dy} \right) = \frac{d}{dx} \left(\frac{dy}{dx} \right)^{-1} \frac{dx}{dy}$$

$$\Rightarrow \frac{d^2x}{dy^2} = -\left(\frac{d^2y}{dx^2} \right) \left(\frac{dy}{dx} \right)^{-2} \left(\frac{dx}{dy} \right)$$

$$\Rightarrow \left(\frac{d^2x}{dy^2} \right) = -\left(\frac{d^2y}{dx^2} \right) \left(\frac{dy}{dx} \right)^{-3}$$

9. (Wrong) $y = \cos^2 x^2$

$$\frac{dy}{dx} = 2 \cos(x^2) (-\sin x^2) (2x) = -4x \cos x^2 \sin x^2$$

10. (a)

$$f(x) = x^3 + e^x + 3^x + \cot x$$

$$f'(x) = 3x^2 + e^x + 3^x \log 3 - \operatorname{cosec}^2 x$$

11. (a)

$$y = [-\log_e(\log_e x), x > 1]$$

On differentiation,

$$\frac{dy}{dx} = -\frac{d}{dx} (\log(\log x)) = -\frac{1}{\log x} \frac{d}{dx} (\log x) = -\frac{1}{x \log x}$$

12. (d) Refer to solution 8.

13. (a) Let, $u = \log_{10} x$ and $v = \log_x 10$

$$\therefore u = \frac{\log_e x}{\log_e 10}, v = \frac{\log_e 10}{\log_e x}$$

$$\frac{du}{dx} = \frac{1}{\log_e 10} \cdot \frac{1}{x} \text{ [differentiating w.r.t. } x \text{]}$$

$$\text{and } \frac{dv}{dx} = -\log_e 10 \cdot \frac{1}{(\log_e x)^2} \cdot \frac{1}{x}$$

$$\text{Now, } \frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{\frac{1}{\log_e 10} \cdot \frac{1}{x}}{-\log_e 10 \cdot \frac{1}{(\log_e x)^2} \cdot \frac{1}{x}}$$

$$= -\frac{1}{\log_e 10} \cdot \frac{1}{x} \cdot \frac{x}{\log_e 10} (\log_e x)^2 = -\frac{(\log x)^2}{(\log 10)^2}$$



14. (b) $y = \tan^{-1} \left[\frac{3x-x^3}{1-3x^2} \right], \left[-\frac{1}{\sqrt{3}} < x < \frac{1}{\sqrt{3}} \right]$

$$y = \tan^{-1}(\tan 3\theta) = 3\theta$$

Put $x = \tan \theta \Rightarrow \theta = \tan^{-1} x \left[\because \tan 3\theta = \frac{3\tan \theta - \tan^3 \theta}{1 - 3\tan^2 \theta} \right]$

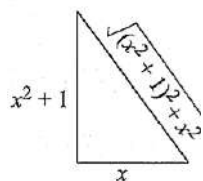
$$y = 3\tan^{-1} x$$

$$\Rightarrow \frac{dy}{dx} = \frac{3}{1+x^2}$$

15. (a) $y = \sin^{-1} \left(\frac{x^2+1}{\sqrt{1+3x^2+x^4}} \right), x > 0$

$$y = \sin^{-1} \left[\frac{x^2+1}{\sqrt{(x^2+1)^2+x^2}} \right]$$

$$y = \tan^{-1} \left(\frac{x^2+1}{x} \right)$$



$$y = \tan^{-1} \left(\frac{x^2+1}{x} \right)$$

$$\frac{dy}{dx} = \frac{1}{1+\left(\frac{x^2+1}{x}\right)^2} \times \frac{x(2x) - (x^2+1)}{x^2} = \frac{x^2-1}{x^4+3x^2+1}$$

16. (d) $x^m y^n = (x+y)^{m+n}$

Taking log both sides,

$$m \ln x + n \ln y = (m+n) \ln(x+y)$$

On differentiating

$$\frac{m}{x} + \frac{n}{y} \frac{dy}{dx} = \frac{m+n}{(x+y)} \left(1 + \frac{dy}{dx} \right)$$

$$\Rightarrow \left(\frac{n}{y} - \frac{m+n}{x+y} \right) \frac{dy}{dx} = \frac{m+n}{x+y} - \frac{m}{x}$$

$$\Rightarrow \left(\frac{n(x+y) - y(m+n)}{y(x+y)} \right) \frac{dy}{dx} = \frac{(m+n)x - m(x+y)}{x(x+y)}$$

$$\Rightarrow \left(\frac{nx - my}{y} \right) \frac{dy}{dx} = \left(\frac{nx - my}{x} \right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x}$$